

VR-Sets. Preprint.

Part VIII. Ontological Position

Vitaly Reznik
2026

VIII.0. Introduction

This part of the preprint develops in detail the ontological position of VR-Sets. The substantive commitments accumulated through Parts I–VII are gathered into a single picture and grounded as a coherent whole. The relation to other well-known positions in the foundations of mathematics — Platonism, formalism, constructivism, structuralism — is also discussed.

Part VIII is the parallel of Part VI of the VR-Numbers preprint (Reznik, 2026), where analogous work was carried out for the ontology of numbers. The emphasis here is transferred to the specifics of a set-theoretic ontology: what changes when "only $\emptyset$ is, all else is doing" is applied to sets.

The part contains no formal theorems. It is a substantive and philosophical exposition of the position for which all the remaining work was undertaken.

VIII.1. "Only $\emptyset$ is, all else is doing"

The slogan running through all three works of the cycle has three levels of meaning, each of which operates in the application to VR-Sets:

Level 1: $\emptyset$ is the sole primitive

In the ontology of VR-Sets there is only one primary entity — $\emptyset$, the empty operationality. All other "things" are actions. This is an ontological decision, not the consequence of any theorem. It is motivated by minimalism: the fewer primitives, the simpler the ontology and the smaller the arbitrary commitments.

The choice of $\emptyset$ specifically as the primitive is likewise motivated. $\emptyset$ is the only entity that can be characterised *purely negatively* (reveals nothing), without requiring a prior content. Any other primitive object would require some "matter," some initial content whose origin would have to be explained. $\emptyset$ is free of this problem by construction.

Level 2: operationality is derived

Operationality — action upon $\emptyset$ and upon the results of prior actions — is not a self-standing kind of being. An operational set is its functionality, but this functionality is not a separate ontological entity; it is a way of acting, related to $\emptyset$.

This distinction is principled (Part I §I.3). Were operationality a "second kind of entity," we would have a dual ontology: $\emptyset$ + operationality. A dual ontology is not minimalist — it has two primitive types that must somehow be related. A derived ontology resolves this

difficulty: operationality "is not" in the proper sense; it "happens" — only its result, visible as a set, is.

A clarifying metaphor: the number 1 is ontologically primary; "to add 1" is not an object but an action; "the addition function" as an object is a derivative speech, useful for mathematics but not a separate kind of number. So too in VR-Sets: $\emptyset$ is primary; operationality is action; "functionality as object" is derivative speech.

Level 3: "doing" as a fundamental modality

The third level is the most philosophical. The slogan asserts not only what is and what is not, but also *in what manner exists* everything that is not $\emptyset$. The manner is "doing," action, process. This is an ontological modality distinct from the classical "being" (Parmenides) and "becoming" (Heraclitus).

In the classical ontology of sets the modality of being predominates: a set *is* with all its properties at the moment of consideration, like a Platonic idea. In VR-Sets the modality of action predominates: a set *acts*, it reveals elements when queried, and this activity is not a property of the set but its essence.

Connection with other philosophical positions: the closest parallel is Whitehead's process ontology (Whitehead, 1929), where the basic entities are events (actual occasions), not objects. VR-Sets is not process philosophy in the full sense (Whitehead's events are temporal; the operationalities of VR-Sets are not), but the methodological kinship is real: in both, action is prior to object.

VIII.2. The ontological commitments of VR-Sets

We collect into one list all the ontological commitments accumulated in the preprint. Each is a consciously accepted decision, not a mathematical necessity.

Commitment 1: the sole primitive is $\emptyset$

All "objects" of VR-Sets are derived from $\emptyset$. No atoms, urelements, individuals, ordered pairs as objects, classes as ontologically new entities. Only $\emptyset$ and operationalities upon it.

Cost: standard constructions requiring pairs (relations, functions via graphs of pairs) are reformulated operationally (see Part III §III.3, Part V §V.5). Such reformulations sometimes lengthen technical proofs; but the ontological gain — the absence of additional primitives — outweighs this.

Commitment 2: only describable objects

All sets of VR-Sets are operationally describable functionalities. Indescribable objects — those without a finite description — do not exist in the ontology. This is the principal divergence from classical ZF, in which the existence of objects without a description is admitted (for instance, a typical element of an uncountable set).

Cost: the countability of the entire universe (Part VI). Uncountable cardinalities, typical uncountable constructions, and paradoxical consequences of full AC on uncountable collections are inaccessible. Standard analysis on $\mathbb{R}$ is restricted to computable reals.

Commitment 3: operationality is derived

Operational functionalities are not a self-standing kind of being alongside $\emptyset$ but derived entities — actions upon $\emptyset$. Ontological status belongs only to $\emptyset$; operationality is in the sense of describedness, not in the sense of primary ontological givenness.

Cost: attentiveness in formulation is required. To say "there exists a set A" in the full ontological sense is incorrect: only $\emptyset$ exists, while A is a describable action upon $\emptyset$. This reformulation adds philosophical discipline but does not alter mathematical claims.

Commitment 4: membership is reference, not containment

$x \in y$ is read as "the functionality y reveals x in one of its responses," that is, as the reference of y to x. Not as "x is physically located inside y." This dissolves the intuitive paradoxes of self-reference (Part I §I.4): $A \in A$ is not "A is located inside itself" but "the functionality A returns A in one of its responses."

Cost: a rejection of the container metaphor of a set, to which classical thought is accustomed. Worth it: the container metaphor is the source of most intuitive difficulties in the foundations of set theory.

Commitment 5: two modes as an explicit ontological choice

The admission or prohibition of self-references (Quine atoms, cycles) is not a mathematical necessity but an ontological choice: ZFC-mode or ZFA-mode. Both modes are legitimate; the choice is made explicitly and depending on the goals of application (Part IV).

Cost: the illusion of a "uniquely correct" set theory is lost. This is not a cost but a clarity: different applications require different frames, and explicit acknowledgement of this is a methodological advantage.

Commitment 6: hierarchy of truth through applications of t

Truth for VR-Sets is not expressible inside VR-Sets in full generality (Tarski's theorem). The hierarchy of applications of the successor operator t gives levels at each of which truth for the previous is expressible, but not for the current level (Part VII).

Cost: there is no single "above-all-levels" truth predicate. This is a commitment shared with all sufficiently rich formal systems by Tarski's theorem; it is not specific to VR-Sets.

VIII.3. Relation to classical positions

We briefly discuss how VR-Sets relates to the principal positions in the philosophy of the foundations of mathematics.

Platonism

Platonism (Gödel, Quine, the early Goodman) asserts that mathematical objects exist independently of our knowledge of them; mathematical cognition is discovery, not construction. Cantorian set theory in its modern form is in large part shaped by Platonist intuitions: the entire hierarchy V is a "really existing" universe, and our theories describe it more or less adequately.

VR-Sets is categorically non-Platonist. There are no sets in VR-Sets existing independently of description. "Only $\emptyset$ is" is the ontological minimum, excluding the entire classical Platonic hierarchy. Mathematical objects not operationally describable are absent in the ontology of VR-Sets — not because we have "not discovered them," but because there is no place for them in the ontology.

Formalism

Formalism (Hilbert) asserts that mathematics is the manipulation of formal symbols without interpretation; the "meaning" of mathematical statements is their derivability in a formal system. Truth in formalism reduces to provability.

VR-Sets is closer to formalism than to Platonism, but does not reduce to it. Operational descriptions are not "symbols without meaning" but instructions for functionalities, with an interpretation through action. The difference: formalism regards mathematics as a game of symbols; VR-Sets, as an ontology of actions for which symbols are merely a way of recording. The action is real (as a process), even if expressed through a formal description.

Constructivism

Constructivism (Brouwer, Heyting, Bishop) asserts that mathematical objects must be constructively given; existence is construction. In its radical form (intuitionism), the law of excluded middle is rejected.

VR-Sets is ontologically close to constructivism: all objects must be operationally given (= constructively built through description). However, VR-Sets does not reject classical logic. The law of excluded middle may be used in VR-Sets, since operational definiteness is already built into the closure principle: if a functionality is operationally defined, it gives a definite response; "the third" is ontologically absent.

The difference: intuitionism rejects classical logic ontologically (there is no objective truth, only justification); VR-Sets retains classical logic ontologically (objects are defined, truth is locally definable) but restricts the ontology to describable objects.

Structuralism

Structuralism (Benacerraf, Shapiro, Resnik) asserts that mathematical objects are positions in structures, and structure matters, not the nature of objects. The natural number 3 is not "a particular object" but "the third position in any structure of natural numbers."

VR-Sets is compatible with structuralism and in a sense refines it. Operational sets are defined through *their behaviour* (what they reveal upon queries), not through "nature."

Isomorphic functionalities are precisely one and the same set by extensionality $\equiv$. This accords with the structuralist intuition: structure matters, not matter.

However, VR-Sets is a more ontological position than structuralism: it does not merely say "matter does not matter" but specifies concretely that only $\emptyset$ is ontologically primary and the rest is action. Structuralism leaves open the question "what then specifically is?"; VR-Sets answers: $\emptyset$ is, action happens.

The position of VR-Sets as "operationalism"

A suitable name for the ontological position of VR-Sets is *operationalism*: mathematical objects are operational actions upon the minimal ontological primitive $\emptyset$. This is:

- **not Platonism** — the ontology contains no objects independent of description;
- **not pure formalism** — symbols are interpreted through actions, not left without meaning;
- **compatible with constructivism** in ontology but without abandoning classical logic;
- **a refinement of structuralism**, specifying that "structure" is a pattern of actions upon $\emptyset$.

This position does not claim to be "uniquely correct." It is a coherent alternative, with its own advantages (ontological minimalism, the absence of paradoxes of uncountability) and limitations (a countable universe, the absence of classical ZFC in its full extent).

A reminder on the status. The discussion in this part has positioned VR-Sets relative to classical ZFC, classical AFA, and other foundations. The *technical* relation between the ZFC-mode and a countable model of classical ZFC (and parallel for the ZFA-mode and AFA) is the content of Conjectures IV.1 and IV.2 and remains open; this is recorded in §IX.1 as Questions 1 and 2. The philosophical positioning of VR-Sets as operationalism does not depend on the resolution of these conjectures: the ontology stands on its own primitives and commitments, whether or not the technical equivalence with a countable model of the classical theory is established.

VIII.4. Applicability: for what VR-Sets is suited

Closing the discussion, we fix which mathematical applications are natural for VR-Sets and which are not.

Naturally suited

- **All of computable mathematics**: algorithms, recursive functions, formal languages, complexity theory, programming.
- **Constructive mathematics** in a broad sense, including reverse mathematics and predicative analysis.
- **Computable analysis**: analysis on computable reals and other computable structures.
- **Discrete mathematics**: graph theory, combinatorics, algebra over countable carriers.
- **Logic and model theory** for countable structures; formal verification of programs.

- **Non-well-founded sets** (through the ZFA-mode): modelling of circular phenomena in semantics, theory of processes, situational semantics of natural language.

Not directly suited

- **Standard analysis** on the classical uncountable $\mathbb{R}$ with the full theory of Lebesgue measure. The analogue in VR-Sets is analysis on $\mathbb{R}_{\text{VR-Sets}}$ of computable reals, which covers most practical applications but differs formally.
- **The theory of uncountable cardinalities:** $\aleph_i$, uncountable ordinals, the continuum hypothesis. In VR-Sets these notions have no direct analogue.
- **Parts of classical functional analysis** relying on uncountable bases, Zorn's lemma for uncountable indices, the existence of ultrafilters. Constructive analogues exist, but not all classical results carry over.
- **Paradoxical consequences of AC:** Banach–Tarski and the like. They are absent by construction (Part VI §VI.5), which is for most purposes an advantage, but if for some reason they are required, VR-Sets is not suited.

The substantive position

VR-Sets is suited for mathematics in which *every object may be exhibited* — operationally, algorithmically, through a description. This is the greater part of substantive mathematics, including practically everything used in informatics, engineering, physics. VR-Sets is not suited for mathematics making essential use of non-exhibitable objects — typical elements of uncountable collections, non-measurable sets, equivalence classes with an uncountable number of representatives. This second area of mathematics exists and is fruitful; VR-Sets does not propose an alternative to it.

Accordingly, the choice between VR-Sets and classical ZFC is not mathematical and not philosophical in the narrow sense, but *methodological*: what kind of mathematics do we intend to pursue? For ontologically transparent, operational, computable mathematics, VR-Sets provides a coherent and minimalist frame. For mathematics requiring classical uncountability, ZFC or its extensions are needed.

VIII.5. Summary

In this part:

- (1) The slogan "only $\emptyset$ is, all else is doing" has been unfolded on three levels of meaning: $\emptyset$ as the sole primitive, operationality as derivative, "doing" as a fundamental modality (§VIII.1).
- (2) Six ontological commitments of VR-Sets have been fixed, with the cost of each indicated (§VIII.2).
- (3) The position of VR-Sets has been related to Platonism (incompatible), formalism (close, but not reducible), constructivism (close ontologically but retaining classical logic), structuralism (compatible, refines). A suitable name for the position is operationalism (§VIII.3).

(4) The applicability of VR-Sets has been described: naturally suited for all of computable and constructive mathematics; not directly suited for the classical theory of uncountable cardinalities (§VIII.4).

Methodological summing-up. VR-Sets is not "better" than classical ZFC and not "worse." It is a different ontological option, with its own area of applicability and its own advantages. The principal advantage is ontological transparency: each object is given explicitly, each existence is grounded through description, and no ontological assumption beyond operationality is made. The principal cost is the countability of the universe, excluding part of classical mathematics.

What follows. Part IX, the final part of the preprint, fixes the open questions: technical (formalisation in Lean, proofs of the equivalence conjectures, machine verification) and substantive (connections with topoi, homotopy type theory, extensions to Morse–Kelley and NBG).